

Dual ordinal spectral ranking of journals and institutions

DOMINGO DOCAMPO AND LAWRENCE CRAM

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Abstract

We develop a dual spectral ranking of journal and institution units that determines influence-per-work scores via the block citation matrix. A binary block-mask selects among four bibliometrically distinct referencing structures, nesting single-unit rankings (such as the journal eigenvector ranking) alongside two forms of journal-institution coupling. Applied to the economics and commerce journal literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: spectral ranking, dual journal-institution prestige, Katz-Hubbell resolvent

1 Introduction

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval and exploration. Institutions fund individual researchers, implement training, oversee compliance, establish research organisational units and maintain physical and information infrastructure. Although the status hierarchy that links journals and institutions is a pervasive aspect of the research system (Merton, 1973; McNamee and Willis, 1994; Martin, 1998), bibliometric studies that explore the joint hierarchy of journals and institutions are rare. This paper proposes a joint ordinal spectral ranking of journals and institutions derived from bibliometric data. The method helps to reveal how status is formed and maintained. We intend to apply the approach to study the relations between the status of journals and institutions, and researcher esteem in later work.

The introduction of the science citation index (Garfield, 1964) opened a new way to investigate connections between journals and institutions. Citation-based journal ranking in particular became a useful means to manage collections and to select outlets for publications. On the other hand, the misapplication of the journal impact factor (JIF) as proxy for the importance of research works published in that journal fostered a still-unresolved tension in bibliometric evaluation (Garfield, 1979; MacRoberts and MacRoberts, 2018). Perhaps due to the complexity of their disciplines and missions, an institution impact factor (IIF) has not emerged. Rather, institutional rankings derived from bibliometric indicators of works, authors or journals are often composed with other indicators to form an institutional score. This is the approach in global rankings such as QS, Times Higher Education (THE) and Shanghai Ranking (ARWU), the academic Leiden and SCImago Institutions rankings, as well as many national performance-based research funding schemes (Zacharewicz et al, 2019).

The prospects for using a citation index to explore research importance were considered in an early work by Narin (1976), Pinski and Narin (1976), and Narin et al (1976) who proposed a recursive method to construct influence weights of journals within a disciplinary field. To obtain influence weights for institutions, Narin summed the institutional publications in a field weighted by the respective journal influence score. Over a sample of several dozen universities, the total institution influence scores were well-correlated with survey-based rankings, but also with the number of publications, suggesting that survey respondents combined influence and size. Narin’s work built on earlier studies of social networks and developed into the eigenvector approach now widely used for journal ranking (Bergstrom et al, 2008; Vigna, 2016; Franceschet, 2010). Eigenvector methods have also been applied to author-level influence weightings in the field of social science, and extended by Narin’s approach to ranking institutions and countries (West et al, 2013). An alternative approach iterates between the article-article network and the corresponding article-author network (Ujum et al, 2015; Pal and Ruj, 2015). The formulation by Cao et al (2023) adopts a similar network, with reference-based article-article links, and the related article-institution links, ranked by a novel and effective iteration over random walks. This is one of very few spectral rankings of institutions. The authors note that a limiting

case of this approach corresponds to the eigenvector (pageRank) method: Geller (1978) noted that random walks over Markov chains are closely related to ranking problems.

It is useful to foreground analytical devices adopted in this paper. First, we say that a *reference* is an entry w_2 in the reference list of a work w_1 : w_1 references w_2 and w_2 is referenced by w_1 . Bibliometric indexers establish the connection between w_2 and the entry for w_2 in an index of the *citations* to w_2 (Wouters, 1999) we say w_1 cites w_2 and w_2 is cited by w_1 . The *attention* that a work gives or pays to its references can be fractionated among its sources and institutions, and those fractions distributed among the sources and institutions of its references. Spectral ranking accounts for the weakening and strengthening of w_1 's attention by the attention paid to it by other sources and institutions, then distributed by fractionation. We can also say that a citation signifies the *influence* of w_2 's authors and institutions on w_1 ; like attention, influence can be weakened or strengthened by other citations to w_2 . Second, in this paper we view references, attention, citations and influence through the lens of Social Systems Citation Theory (Tahamtan and Bornmann, 2022, SSCT). We concentrate on the structural features of the network of references and citations, setting aside authorial intent. This approach contrasts with actor-network theory (Latour, 2007, ANT), which foregrounds the intentful roles of heterogeneous agents in assembling citation networks. ANT will be important in our follow-up studies that relate this paper to the referencing activities of individual researchers.

This paper proceeds as follows. In the next section, we construct a framework for computing the transfer of attention or influence among journal and institution units coupled by the reference lists of the works they publish. We then construct a corpus of works in the fields of Economics and Business, forming subsets of the corpus to illustrate the interconnections of units, fields, and time. We present spectral rankings over these subsets, alongside sensitivity and bootstrap studies of some key parameters. We then discuss the results, including a critique of our methods, and close by mentioning some directions for future work.

2 Journal–institution citation networks

We begin by selecting N_s sources $\mathfrak{S}$ in a set of fields, disciplines or categories $\mathfrak{F}$. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $t_c = [t_0^c \dots t_1^c]$ or target years $t_t = [t_0^t \dots t_1^t]$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{r(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of institutions $\mathfrak{U}'$ and authors $\mathfrak{A}'$. In practice many institutions (and sources) publishing only a few works per year can be dropped from the corpus. Let $\bar{w}_u$ be the mean annual census-window work count for institution u and $\bar{w}_s$ the corresponding work count for source s . We determine lower limits τ_U and τ_S by examining the respective frequency distributions and define $\mathfrak{U} \subseteq \mathfrak{U}'$ as the institutions with $\bar{w}_u \geq \tau_U$, and $\mathfrak{S} \subseteq \mathfrak{S}'$ as the sources with $\bar{w}_s \geq \tau_S$, with corresponding author set $\mathfrak{A} \subseteq \mathfrak{A}'$. Works that lose all their authorships under the institution filter, or whose source is dropped by the source filter, are removed from the corpus.

We say that a work w_i in $\mathfrak{W}^c$ transfers (or pays) *attention* to its references r_{ij} . Let $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. Since reference lists have different lengths, we consider two cases for the transferred attention per reference,

$$\rho_i = \begin{cases} 1 & \text{(full reference counting)} \\ \bar{R}/R_i & \text{(fractional reference counting)}. \end{cases} \quad (1)$$

Under full reference counting a work with twice as many references transfers twice as much attention; under fractional reference counting every work transfers the same total attention $\bar{R}$ (Zitt and Small, 2008).

A work's attention can be conferred on or allocated to its retained source and institutions. The corpus parameter χ sets the division of conferred attention between the source (χ) and institutions ($1 - \chi$). While at first sight the choice $\chi = 0.5$ appears sound, we see below that $\chi = N_S/(N_S + N_I)$ arises naturally once all works have been accumulated. We consider two models for conferral of a work's attention on its institutions (i.e. the *institution weight*)

$$\omega_{iu} = \begin{cases} \sum_{\ell: u \in \Pi_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, & \text{(author-fractional)} \\ \frac{1}{n_i}, & \text{(institution-fractional)} \end{cases} \quad (2)$$

where $n_i = |\bigcup_{\ell} \square_{i\ell}|$ is the number of institutions of work w_i , and ℓ indexes the authors of work w_i , a_i is their count, and $u_{i\ell}$ the number of institutions of author ℓ , so that $\sum_u \omega_{iu} = 1$ under author-fractional counting (Waltman and van Eck, 2015). The decision to re-weight fractional counts after any institution pruning is a trade-off between boosting a work’s remaining institutions and diminishing the work’s total contribution.

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \dots, N_s$ and institutions indices $N_s + 1, \dots, N$. Each reference $r(i, j) \in \mathfrak{R}$ carries attention from the units associated with w_i to the units associated with w_j , across four unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (2). The attention carried by each reference is distributed across unit pairs and accumulated as four block matrices over $\mathfrak{R}$ via rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \rho_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}, \quad (3)$$

where $\mathbf{b}_i^X$ distributes attention from the citing side across units of type X , and $\mathbf{b}_j^Y$ distributes it to the cited side across units of type Y . Each entry $[\mathbf{C}_{XY}]_{pq}$ is therefore proportional to the ρ_i -weighted attention flowing from unit p of type X to unit q of type Y across all references in $\mathfrak{R}$.

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \rho) = \begin{bmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{bmatrix}, \quad (4)$$

where the binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$ is bibliometrically meaningful for source-only (1000), institution-only (0001), bipartite (0110), and full-joint (1111) combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The χ scaling preserves total attention flow across all block combinations, since $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$.

Note that the corpus excludes (a) census work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

Ordinal spectral ranking. The attention of a unit is amplified or diminished by the attention that it receives from the units connected to it via the citation matrix. This leads to an eigenvector equation in which the attention of each unit is defined recursively as the weighted sum of the attention of the units that reference it, with its value established by the convergence of that recursion. The process of finding the fixed point is *spectral ranking* (a verb); the resulting vector is the *spectral ranking* (a noun). The vector can be read from either side of the transaction — from the referencing side as attention, or from the cited side as influence — just as $\mathbf{C}$ can be called a reference matrix or a citation matrix depending on which side one stands.

Let $r_p = \sum_q C_{pq}$ be the sum of fractionated references given by unit p and $\mathbf{D}_r = \text{diag}(r_p)$ the corresponding diagonal matrix. Row-normalising $\mathbf{C}$ to obtain $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$ ensures that each unit’s fractionated references are equally valued regardless of how many references it makes: a unit with many references exerts no more attention per reference than one with few (this consideration obtains when $\rho_i = 1$). Reference intensity variations are removed at this stage and reintroduced only at the end via the influence per work scaling (7). If $r_p = 0$ (a dangling unit with no outgoing references) the corresponding row of $\mathbf{H}$ is defined to be zero.

The three unipartite masks—source-only (1000), institution-only (0001), and full-joint (1111)—each yield a square $\mathbf{H}$ to which the following spectral ranking theory applies directly. The spectral ranking $\boldsymbol{\pi}^*$ (see Vigna, 2016, for an overview) is the unique positive solution to the eigenvector equation,

$$(\mathbf{I} - \mathbf{H}^\top) \boldsymbol{\pi}^* = \mathbf{0}, \quad \mathbf{1}^\top \boldsymbol{\pi}^* = 1, \quad \boldsymbol{\pi}^* \geq 0, \quad (5)$$

if $\mathbf{H}$ is primitive (irreducible and aperiodic), as guaranteed by the Perron–Frobenius theorem. Our numerical implementation tests primitivity for every $\mathbf{H}$ as explained below and has not encountered an imprimitive case. Geller (1978) showed that $\boldsymbol{\pi}^*$ is the stationary distribution of the Markov chain with transition matrix

$\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion $r_p / \sum_q r_q$.

Katz (1953) proposed attenuating attention along walks of length k by a factor α^k , $\alpha \in (0, 1)$, so that distant attention counts for less than proximate ones. Summing attenuated walks of all lengths and endowing every unit with a uniform initial score of 1 yields the Katz resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = \mathbf{1}.$$

Since $\mathbf{H}$ is row-stochastic, $(\mathbf{I} - \alpha \mathbf{H}^\top)$ is invertible for any $\alpha \in (0, 1)$.

Hubbell (1965) generalised this by replacing the uniform endowment $\mathbf{1}$ with an informative prior $\boldsymbol{\mu}$ over units, representing exogenous information about standing independent of citation structure,

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = \boldsymbol{\mu}.$$

We use the uniform prior $\mu_p = 1/N_{\mathfrak{P}}$ for $p \in \mathfrak{P}$. Note that the intermediate case $(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = \mathbf{0}$ with $\alpha \neq 1$ admits only the trivial solution $\boldsymbol{\pi} = \mathbf{0}$ and is not useful as a ranking.

Combining attention attenuation and prior information yields the general resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = (1 - \alpha) \boldsymbol{\mu}, \quad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \quad (6)$$

which recovers $\boldsymbol{\pi}^*$ in the limit $\alpha \rightarrow 1$ when $\mathbf{H}$ is primitive. While α and $\boldsymbol{\mu}$ are introduced here for their substantive interpretations—attention attenuation and exogenous prior respectively—they have the additional effect of ensuring a unique solution to (6) even when $\mathbf{H}$ is not primitive: $\alpha < 1$ makes $(\mathbf{I} - \alpha \mathbf{H}^\top)$ invertible, and a nonzero $\boldsymbol{\mu}$ lifts the equation off the homogeneous system. If $\mathbf{C}$ were a binary matrix, this would be the PageRank spectral ranking (Brin and Page, 1998)

The spectral ranking $\boldsymbol{\pi}$ reflects the total attention or influence of each unit and depends on unit size. The *influence per work* (Palacios-Huerta and Volij, 2004),

$$\mathbf{v} = A \mathbf{A}^{-1} \boldsymbol{\pi}, \quad \mathbf{A} = \text{diag}(a_p), \quad A = \mathbf{a}^\top \mathbf{1} = \sum_p a_p, \quad (7)$$

removes this dependence, where the unit work count $a_p = |\{i : w_i \in p\}|$ where p is a specific source or institution, and $\boldsymbol{\pi}$ denotes the solution to (6). The scalar A normalises $\mathbf{v}$ so that its a_p -weighted mean equals one: $\mathbf{a}^\top \mathbf{v} / A = \mathbf{1}^\top \boldsymbol{\pi} = 1$. A unit with $v_p = 1$ therefore has the corpus-average influence per work and $v_p = 2$ indicates twice the average influence per work. Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004).

Bipartite ranking. With $\mathbf{m} = (0110)$, equation (4) yields block off-diagonal matrices $\mathbf{C}$ and $\mathbf{H}$, forming a directed bipartite network between sources (S) and institutions (I). This is the mask used as the baseline model in Section 4. Partitioning rows and columns into the S and I blocks, the stochastic matrix takes the form

$$\mathbf{H} = \begin{pmatrix} \mathbf{0} & \mathbf{H}_{SI} \\ \mathbf{H}_{IS} & \mathbf{0} \end{pmatrix}, \quad (8)$$

where $\mathbf{H}_{SI}$ distributes a source’s attention across institutions and $\mathbf{H}_{IS}$ does the reverse. Because every path alternates $S \rightarrow I \rightarrow S \rightarrow \dots$, this structure makes $\mathbf{H}$ periodic with period 2.

Squaring $\mathbf{H}$ collapses two-step walks to separate node types, giving one-mode relations

$$\mathbf{H}^2 = \begin{pmatrix} \mathbf{M}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_I \end{pmatrix}, \quad \mathbf{M}_S = \mathbf{H}_{SI} \mathbf{H}_{IS}, \quad \mathbf{M}_I = \mathbf{H}_{IS} \mathbf{H}_{SI}.$$

Both $\mathbf{M}_S$ and $\mathbf{M}_I$ are row-stochastic products of row-stochastic matrices.

To make the bipartite mode comparable with the unipartite SS and II modes at the same α , single steps SI or IS should be damped by $\sqrt{\alpha}$, so the round-trip incurs attenuation $\sqrt{\alpha} \cdot \sqrt{\alpha} = \alpha$. Substituting $\sqrt{\alpha}$ for α in equation (5), partitioning $\boldsymbol{\pi} = (\boldsymbol{\pi}_S, \boldsymbol{\pi}_I)$, and eliminating $\boldsymbol{\pi}_I$ yields the bipartite resolvent for sources,

$$(\mathbf{I} - \alpha \mathbf{M}_S^\top) \boldsymbol{\pi}_S = (1 - \sqrt{\alpha}) (\boldsymbol{\mu}_S + \sqrt{\alpha} \mathbf{H}_{IS}^\top \boldsymbol{\mu}_I), \quad (9)$$

Note that when $\alpha < 1, \mu > 0$, the right-hand side links institutions to sources via $\mathbf{H}_{IS}^\top$. Institution ranking can be computed from source ranking by the matrix–vector product,

$$\boldsymbol{\pi}_I = \sqrt{\alpha} \mathbf{H}_{SI}^\top \boldsymbol{\pi}_S + (1 - \sqrt{\alpha}) \boldsymbol{\mu}_I. \quad (10)$$

Computation. Defining the modified matrix

$$\tilde{\mathbf{H}} = \alpha \mathbf{H} + (1 - \alpha) \boldsymbol{\mu} \mathbf{1}^\top, \quad (11)$$

the resolvent $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$ is equivalent to equation (6). This is the general form of our numerical problem, although most of the results reported below use $\alpha = 1$ and $\boldsymbol{\mu} = 0$ so $\tilde{\mathbf{H}} = \mathbf{H}$.

As explained below, we construct many realisations of $\tilde{\mathbf{H}}$ while testing different models for unit pairings, subfields and other parameters, and to run our bootstrap investigations. These realisations correspond to a unit-unit citation network with one giant strongly connected component (SCC) and some very small, usually singleton, components that are sources or institutions disjoint with the particular network under investigation. In these cases we retain only the SCC. For some units in the SCC, there may be no outgoing references, and for others, no incoming references. We exclude these units too, ensuring that the residual $\tilde{\mathbf{H}}$ is highly likely to be row-stochastic and primitive, with a single dominate eigenvector.

$\tilde{\mathbf{H}}$ is a sparse matrix constructed from the respective filtered and weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$ and stored in compressed sparse row (CSR) format. We have explored two approaches to applying the resolvent. The Python scipy function `scipy.sparse.linalg.eigs` efficiently provides all eigenpairs, allowing us to review the primitivity condition. The power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \quad (12)$$

converges to $\boldsymbol{\pi}$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$. In either approach, the per-work scaling equation (7) is applied after $\boldsymbol{\pi}$ has been computed.

3 Corpus selection

OpenAlex (Priem et al, 2022) provides our bibliometric data. Corpus selection starts with a list of eligible sources having four features: an OpenAlex source identifier for corpus extraction, evidence from one of our journal registries that the source publishes economics or business research, a sufficient number of published works per year, and relatively high OpenAlex source topic density excluding cross-disciplinary journals.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is confined to economics and business so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having all FoR codes falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit ‘division’ codes and all four-digit ‘group’ codes (3501–3509, 3801–3803, 3899). After filtering, 1,602 journals remain. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance. After filtering, 1,413 journals remain.

Every journal in JQL, SJL, and JCR has at least one ISSN, and may have several. These registry ISSNs are joined independently to the February 2026 OpenAlex sources snapshot using its list of ISSNs as the matching key and selecting only the ‘journal’ source-type. Journal titles are examined to identify around 30 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS*. For OAS* sources we extract work and authorship data revealing some units that can be excluded because they publish very few works. To find a suitable threshold for excluding these units, we examine source and institution retention curves (Fig. 1), selecting thresholds of $\tau_U = 20$ works/institution/year and $\tau_S = 20$ works/source/year. We designate the residual set of sources OAS.

Using units that satisfy these thresholds, we extract a full corpus of works, authorships and references retaining article and review types with publication years from 2000 to 2024, excluding works that are paratext or retracted, or without an authorship entity, and compute ω_{iu} for each work. The characteristics of the full corpus are exhibited in Table 2.

Our sources cover the economics and business subfields, and it is of interest to compare spectral rankings for both. The three registries include one or more subfield indicators for each source, and OpenAlex sources have topic scores that indicate the relative intensity of business and economics subfields. We classify sources

Table 1: Matching of the three register-based sources to OpenAlex source identifiers. Overlaps count shared matched entries; OAS* is the union across all three registries.

Register	Sources	Matched	Match rate	Unmatched
JQL	842	836	99.3%	6
MJL	1,413	1,392	98.5%	21
SJL	1,602	1,370	85.5%	232
Overlap: MJL $\cap$ SJL				782 sources
Overlap: MJL $\cap$ JQL				567 sources
Overlap: SJL $\cap$ JQL				422 sources
Overlap: MJL $\cap$ SJL $\cap$ JQL				393 sources
(JQL $\cup$ MJL $\cup$ SJL) $\cap$ OA (OAS*)				2,359 sources

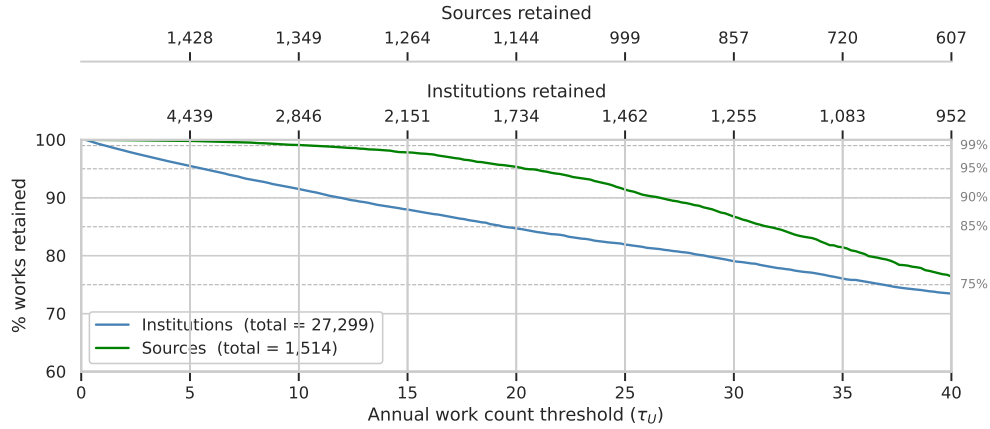


Figure 1: Institution and source retention curves

Table 2: Corpus features by year ($\tau_U > 20$). Upper panel: unique-item counts. Lower panel: D_1 and D_9 deciles of per-entity distributions. The 2000–24 column pools the full period.

Quantity	2000		2024		2000–24	
	count		count		count	
Works	29,920		107,785		1,695,241	
Sources	1,050		1,905		2,310	
Institutions	1,294		1,506		1,508	
Reference counts (out-degree)	10,805		2,511,086		25,172,159	
Citation counts (in-degree)	975,803		50,595		25,247,139	
Quantity	D_1	D_9	D_1	D_9	D_1	D_9
Works per source	4.0	55.0	5.0	104.0	37.0	1588.1
Institutions per work	1.0	3.0	1.0	5.0	1.0	4.0
References per work (out-degree)	1.0	3.0	3.0	58.0	2.0	42.0
Citations per work (in-degree)	2.0	84.0	1.0	4.0	1.0	42.0

as ‘E’ or ‘B’ when the number of subfield indicators summed across the registries and OA exceeds a specified threshold, ‘A’ when both subfields are present above the threshold, and ‘X’ for sources that don’t reach the threshold in either E or B. While sources are readily related to subfields, institutions generally publish substantial numbers of works in both subfields. Therefore, rather than use categorical classification, we classify institutions based on the ratio of works in sources across the EBAX categories.

4 Results

Outline. Tables 3–5 define and enumerate the parameters; all are consolidated here and not repeated elsewhere. The *baseline* is $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$, $\mu = 0$ — the pure eigenvector equation (5) on the bipartite network. Modes $\mathbf{m} = 1000$ (SS) and 0001 (II) are the unipartite analogues of standard journal and institution eigenvector methods; 1111 adds all paths between the two unit types; 0110 is the bipartite focus of this paper. Phase 1 fixes $\alpha = 1$, $\mu = 0$ and varies the corpus (F) and mode ($\mathbf{m}$); Phase 2 varies α , ρ , and τ ; Phase 4 varies $t_x \in \{1, 2, 3, 4, 5\}$.

Table 3: Model parameters θ .

Symbol	Name	Type	Role
<i>Corpus construction</i>			
$F \in \{E, B, A\}$	field subset	category	restrict $\mathfrak{S}$ to Economics (E), Business (B), or all OAS (A); see Table 4
$t_x \in \{1, \dots, 5\}$	time window	integer	census and target year ranges; see Table 4
τ_U	institution threshold	integer	minimum mean annual works to retain an institution
τ_S	source threshold	integer	minimum mean annual works to retain a source
<i>Matrix construction</i>			
$\rho \in \{0, 1\}$	reference normalisation	binary	0: a work contributes $\bar{R}$ units 1: a work contributes R_i units
$\mathbf{m} \in \{0, 1\}^4$	block mask	binary 4-vector	select active unit-type pairs (SS , SI , IS , II)
$\chi \in [0, 1]$	mixing weight	continuous	source–institution balance within each work
<i>Ranking</i>			
$\alpha \in [0, 1)$	damping factor	continuous	controls propagation through citation network
μ	prior weight	continuous	exogenous prestige prior; $\mu = 0$ = pure spectral

Phase 1: Sources, institutions and fields. Figure 2 shows the prestige-per-work distribution for the baseline and for the field subsets $F = E$ and $F = B$, with the x -axis locked to the baseline rank so that each unit sits at its baseline position and absent units are omitted from that series. Sources at the Economics–Business boundary lose prestige flow from one side under each restriction; the institution panel reflects the same effect through the authorship mapping.

The field shifts in Figure 2 reflect community structure in the citation network: sources cluster into Economics and Business communities that preferentially cite within their own field, and this influences rankings because prestige propagates more readily within a community than across it.

The second eigenpair (λ_2, ϕ_2) of $\mathbf{H}^\top$ characterises this structure precisely. For general α and uniform prior $\mu_p = 1/N$, the solution to (6) has the spectral expansion

$$\boldsymbol{\pi} = (1 - \alpha) \sum_{i \geq 1} \frac{\phi_i^\top \boldsymbol{\mu}}{1 - \alpha \lambda_i} u_i, \quad (13)$$

Table 4: Corpus construction parameters: field subsets (F) and time windows (t_x).

Panel A: Field subsets (F)

F Sources included in $\mathfrak{S}$

E Topic counts exclusively in Field 14 (Economics, Econometrics and Finance)

B Topic counts exclusively in Field 20 (Business, Management and Accounting)

A All OAS sources; topic counts in Fields 14 and/or 20 (*baseline*)

Panel B: Time windows (t_x)

t_x	Window $[t_0^c, t_1^c] = [t_0^t, t_1^t]$	Label
1	$t_{00} = [2000, 2004]$	Symmetric 5-year
2	$t_{05} = [2005, 2009]$	Symmetric 5-year
3	$t_{10} = [2010, 2014]$	Symmetric 5-year
4	$t_{15} = [2015, 2019]$	Symmetric 5-year
5	$t_{20} = [2020, 2024]$	Symmetric 5-year (<i>baseline</i>)

Table 5: Parameters and values explored. Baseline values in italics.

Symbol	Name	Baseline	Values explored
<i>Corpus construction</i>			
t_x	Time window	<i>5</i>	1, 2, 3, 4, 5
F	Field subset	<i>A</i>	<i>E</i> , <i>B</i> , <i>A</i>
τ_U	Institution threshold	<i>20</i>	<i>20</i> , 40
τ_S	Source threshold	<i>20</i>	<i>20</i> , 40
<i>Matrix construction</i>			
ρ	Reference normalisation	<i>0 (fixed)</i>	0 (fixed), 1 (full)
$\mathbf{m}$	Block mask	<i>0110</i>	1000, 0001, <i>0110</i> , 1111
χ	Mixing weight	0.50	0.50, χ^* [full joint only]
<i>Ranking</i>			
α	Damping factor	<i>1.0</i>	0.1, 0.5, <i>1.0</i>
μ	Prior weight	<i>0</i>	0 (pure spectral)

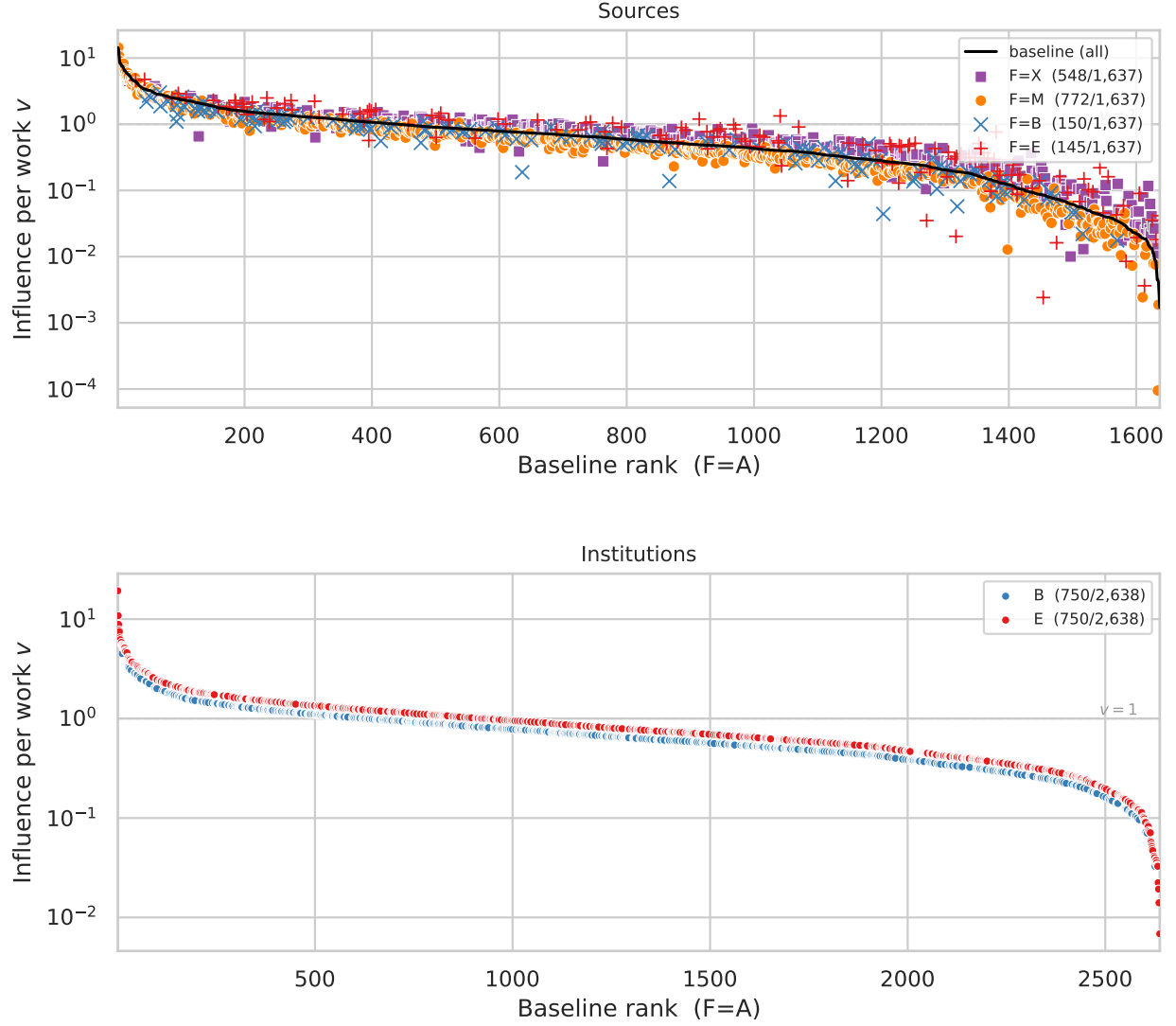


Figure 2: Prestige per work v against baseline rank. Top: sources. Bottom: institutions. Black line: baseline ($F = A$). Red +: $F = E$ ($N_s = [FILL]$). Blue $\times$: $F = B$ ($N_s = [FILL]$). Dashed line: $v = 1$ (corpus average). Parameters: $t_x = 5$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

where (λ_i, u_i, ϕ_i) are the eigenvalues, right eigenvectors, and left eigenvectors of $\mathbf{H}^\top$, ordered $\lambda_1 \geq \lambda_2 \geq \dots$. The $i = 1$ term recovers π^* as $\alpha \rightarrow 1$; the $i = 2$ term is the leading community correction, amplified by $1/(1 - \alpha\lambda_2)$ relative to all higher terms. The spectral gap $g = 1 - \lambda_2$ therefore governs community influence: small g means community membership strongly shapes rankings; large g means rankings are community-neutral. For the SS mode ϕ_2 is expected to separate Economics from Business sources; for II the partition is more diffuse because institutions co-locate faculty from both fields. In the bipartite mode the community operator is the one-mode projection $\mathbf{M}_S = \mathbf{H}_{SI}\mathbf{H}_{IS}$; ϕ_2^S is its second eigenvector and the institution community vector is recovered as $\phi_2^I = \mathbf{H}_{SI}^\top \phi_2^S$ (unit-normalised). Throughout, ϕ_2 is signed so that the unit with the largest work count a_p has $\phi_{2,p} > 0$.

Figure 3 illustrates these eigenvectors by plotting the size-adjusted community measure $\phi_{2,p}/\sqrt{v_p}$ against v_p ; dividing by $\sqrt{v_p}$ removes the normalisation artefact that suppresses $|\phi_{2,p}|$ for low-prestige units.

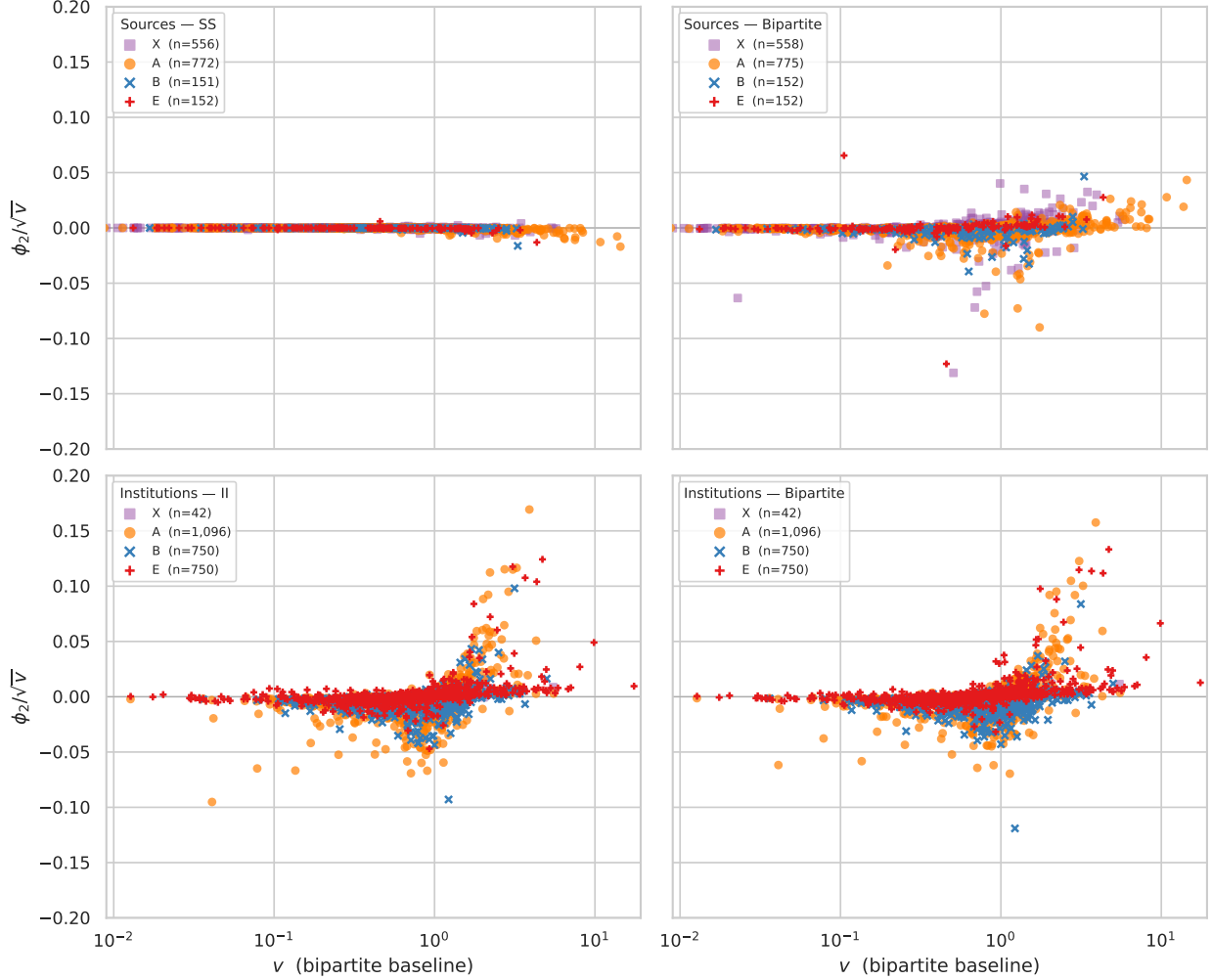


Figure 3: Community measure $\phi_{2,p}/\sqrt{v_p}$ against prestige per work v_p . Top row: sources. Bottom row: institutions. Left column: SS mode $\mathbf{m} = 1000$. Right column: bipartite $\mathbf{m} = 0110$. Sources coloured by field: blue circles = Economics (E); red squares = Business (B); grey crosses = other. Institutions coloured by field-network membership (E-only, B-only, other). x-axis: bipartite baseline v . Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\alpha = 1$.

In the SS source panel (top left), the sign of $\phi_{2,p}/\sqrt{v_p}$ [FILL: cleanly separates Economics (positive) from Business (negative) journals across the full prestige range, with Finance journals concentrated near zero, confirming their role as citation bridges between the two communities]. The bipartite source panel (top

right) shows [FILL: a similar partition / greater mixing], with spectral gaps $g^{SS} = [FILL]$ and $g^{bip} = [FILL]$ indicating that routing citation flow through institutions [FILL: partially dissolves / preserves] the community boundary. The institution panels (bottom row) show a more diffuse partition, consistent with institutions co-locating faculty from both fields; the field-network colouring confirms that E-only institutions [FILL: concentrate on one side / are scattered].

Figure 4 compares prestige-per-work distributions across the four network modes, with the x -axis locked to the bipartite baseline. For sources, modes 1000 (red $\times$) and 1111 at χ^* (green $\times$) are overlaid on the bipartite baseline; open circles and squares on the 1000 series mark Economics and Business sources respectively. For institutions, mode 0001 and 1111 replace the source alternatives.

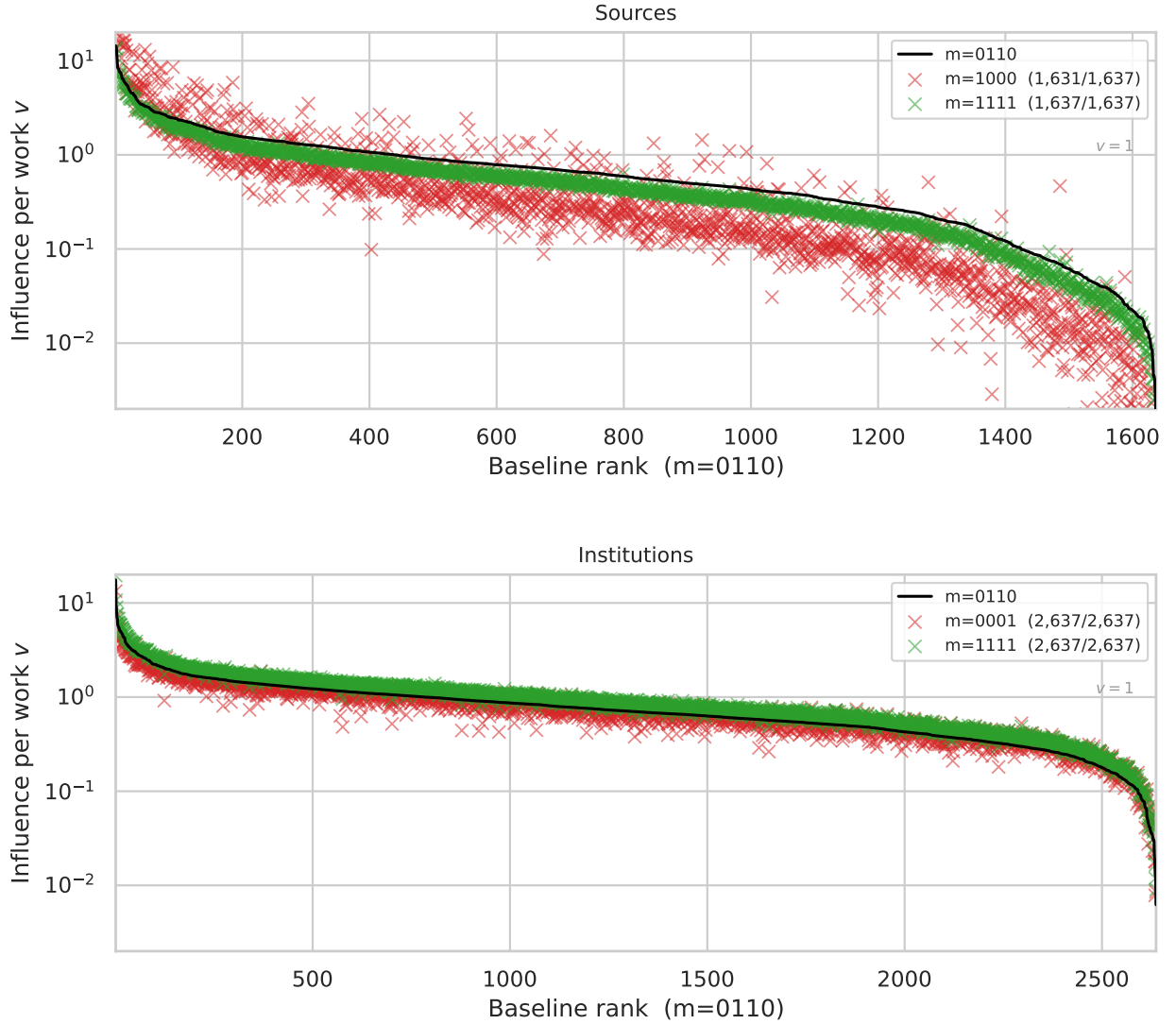


Figure 4: Prestige per work v against bipartite baseline rank. Top: sources. Bottom: institutions. Black line: bipartite baseline $\mathbf{m} = 0110$. Red $\times$: $\mathbf{m} = 1000$ (top) / $\mathbf{m} = 0001$ (bottom). Green $\times$: $\mathbf{m} = 1111$, $\chi^* = [FILL]$. Open circles ($\circ$) and squares ($\square$) overlaid on the 1000 series (top) and 0001 series (bottom) mark Economics and Business units respectively, classified by field-network membership. Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\alpha = 1$.

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Addenda

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